

Correlation - degree of similarity b/w 2 signals

Auto Corr.

$$R_{xx}(n) = \sum_{k=-\infty}^{\infty} x(k) x(k-n)$$

Cross Corr.

$$R_{xy}(n) = \sum_{k=-\infty}^{\infty} x(k) y(k-n)$$

$$k \text{ if } (xy) \rightarrow x \text{ range} \\ n \rightarrow \max_{\substack{\text{max} \\ x}} - \min_y \quad \min_{\substack{\text{min} \\ x}} - \max_y$$

$$Q. \quad x(n) = \{ \overset{-2}{1}, 1, \underset{\uparrow}{0}, 1 \} \quad \Rightarrow x(k)$$

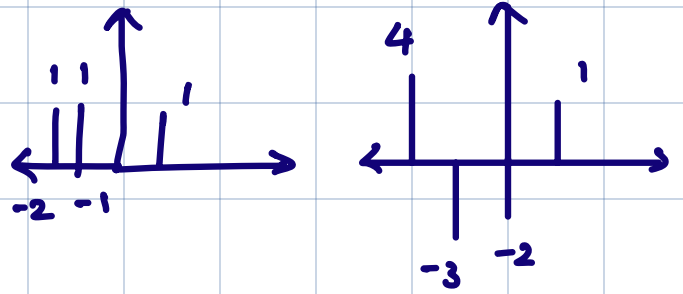
$$y(n) = \{ 4, -3, -2, 1 \} \quad \Rightarrow y(k)$$

$k: -2 \text{ to } 1$

$$n: \max = 1 - (-2) = 3$$

$$\min = -2 - (1) = -3$$

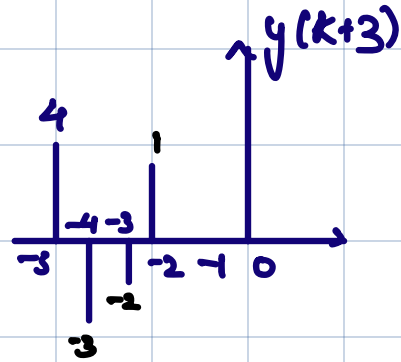
$$R_{xy}(n) = \sum_{k=-\infty}^{\infty} x(k)y(k-n)$$



$$\underline{n = -3}$$

$$R_{xy}(-3) = \sum_{k=-\infty}^{\infty} x(k)y(k+3)$$

$$= 1 \times 1$$

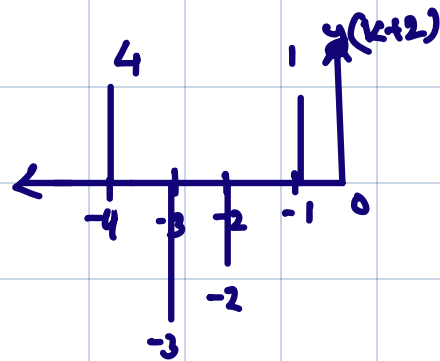


$$n = -2$$

$$R_{xy}(-2) = \sum_{k=-\infty}^{\infty} x(k)y(k+2)$$

$$= -2 \times 1 + 1 \times 1$$

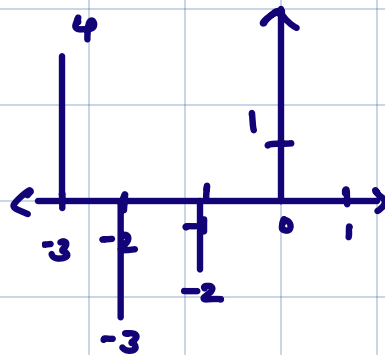
$$= -1$$



$$R_{xy}(-1) = \sum_{k=-\infty}^{\infty} x(k)y(k+1)$$

$$= -3 - 2$$

$$= -5$$

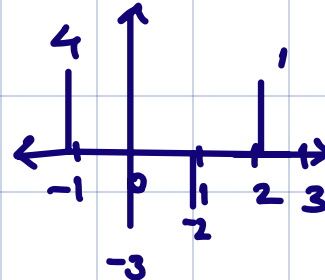


$$R_{xy}(0) = \sum_{k=-\infty}^{\infty} x(k)y(k)$$

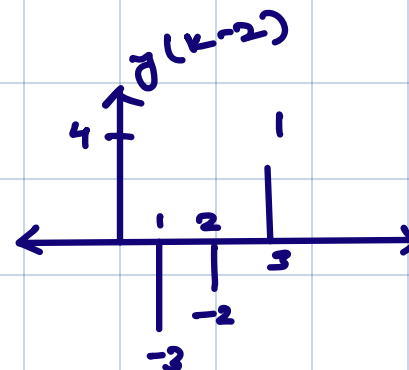
$$= 1 \times 4 + 1 \times -3 + 1 \times 1$$

$$= 2$$

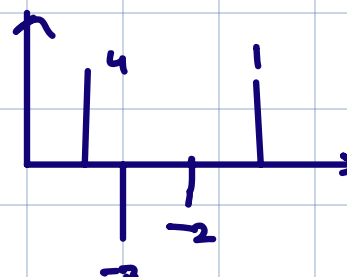
$$\begin{aligned}
 R_{xy}(1) &= \sum x(k) y(k-1) \\
 &= 1 \times 4 + 0 + 1 \times -2 \\
 &= 2
 \end{aligned}$$



$$\begin{aligned}
 R_{xy}(2) &= \sum x(k) y(k-2) \\
 &= 1 \times -3 = -3
 \end{aligned}$$



$$\begin{aligned}
 R_{xy}(3) &= \sum x(k) y(k-3) \\
 &= 1 \times 4 = 4
 \end{aligned}$$



Tabular Form

$\downarrow x(n)$

	1	1	0	1
flip $y(n)$	1	1	0	1
-2	-2	-2	0	-2
-3	-3	-3	0	-3
4	4	4	0	4

1, -2, -5, 2, 2, -3, 4

$$Q. \quad x(n) = \underset{\uparrow}{\overset{0}{1}}, 2, 1, \overset{3}{1} \Rightarrow x(k)$$

$$R_{xx}(n) = \sum_{k=-\infty}^{\infty} x(k) x(k-n)$$

$$k \Rightarrow 0 \text{ to } 3$$

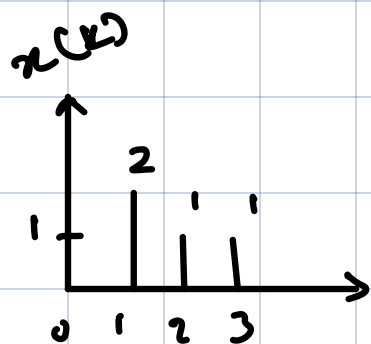
$$n \Rightarrow$$

max

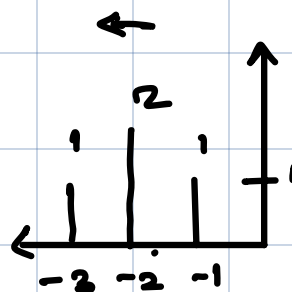
$$3 - 0 = 3$$

min

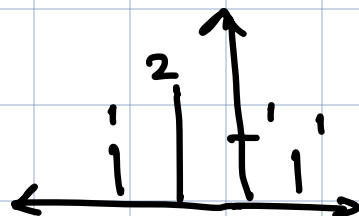
$$0 - 3 = -3$$



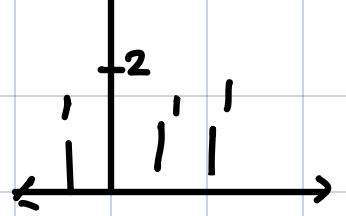
$$\begin{aligned} R_{xx}(-3) &= \sum_{k=-\infty}^{\infty} x(k) x(k+3) \\ &= 1 \end{aligned}$$



$$\begin{aligned} R_{xx}(-2) &= \sum x(k) x(k+2) \\ &= 1 + 2 = 3 \end{aligned}$$

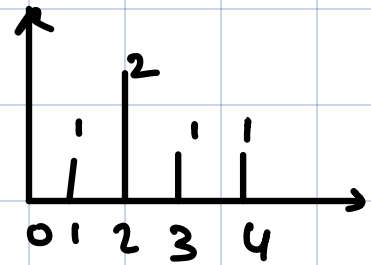


$$\begin{aligned}
 R_{xx}(-1) &= \sum x(k) x(k+1) \\
 &= 2 + 1 + 1 \\
 &= 1 \times 2 + 2 \times 1 + 1 \times 1 = 5
 \end{aligned}$$

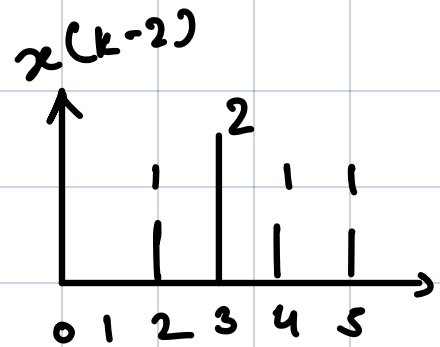


$$\begin{aligned}
 R_{xx}(0) &= \sum x(k) x(k) \\
 &= 1 \times 1 + 2 \times 2 + 1 \times 1 + 1 \times 1 \\
 &= 1 + 4 + 1 + 1 = 7
 \end{aligned}$$

$$\begin{aligned}
 R_{xx}(1) &= \sum x(k) x(k-1) \\
 &= 2 \times 1 + 1 \times 2 + 1 \times 1 \\
 &= 5
 \end{aligned}$$



$$\begin{aligned}
 R_{xx}(2) &= \sum x(k) x(k-2) \\
 &= 1 \times 1 + 1 \times 2 \\
 &= 4
 \end{aligned}$$



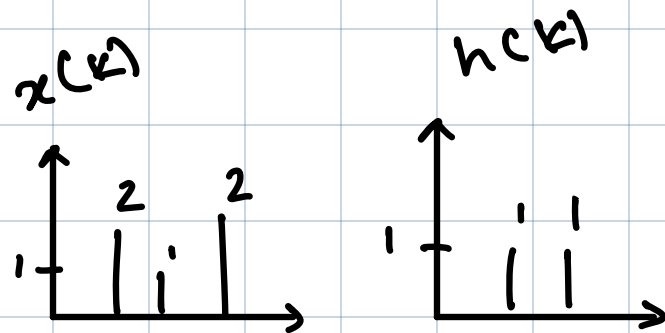
$$\begin{aligned}
 R_{xx}(3) &= \sum x(k) x(k-3) \\
 &= 1
 \end{aligned}$$

Convolution

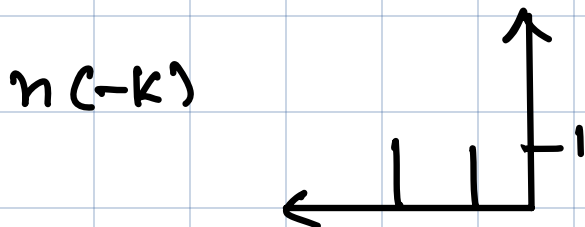
↳ Linear $y(n) = \sum_{k=-\infty}^{\infty} x(k) h(-k+n)$

Q. $x(n) = \{1, 2, 1, 2\}$ $x(k)$
 $h(n) = \{1, 1, 1\}$ $h(k)$

$$\begin{array}{rcl} n \Rightarrow & 0 \rightarrow 3 & \\ & 0 \rightarrow 2 & \\ \hline & 0 \rightarrow 5 & \end{array}$$



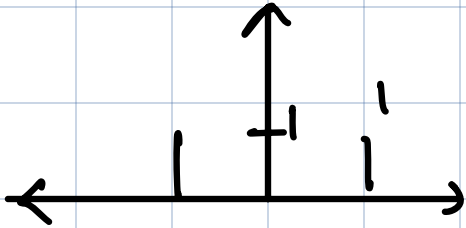
$$y(0) = \sum_{k=-\infty}^{\infty} x(k) h(-k+0)$$



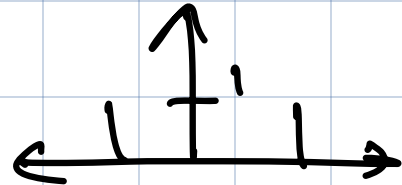
$$y(0) = 1$$

$$y(1) = \sum_{k=-\infty}^{\infty} x(k) h(-k+1)$$

$$h(-k+1)$$

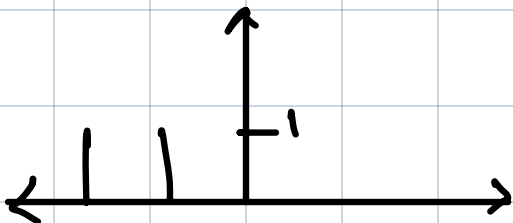


$\Rightarrow$

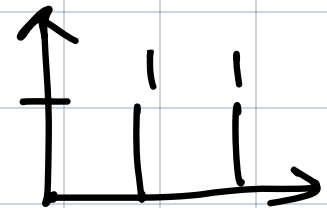


$$y(1) = 1 \times 1 + 2 \times 1 = 3$$

$$y(2) = \sum_{k=-\infty}^{\infty} x(k) h(-k+2)$$

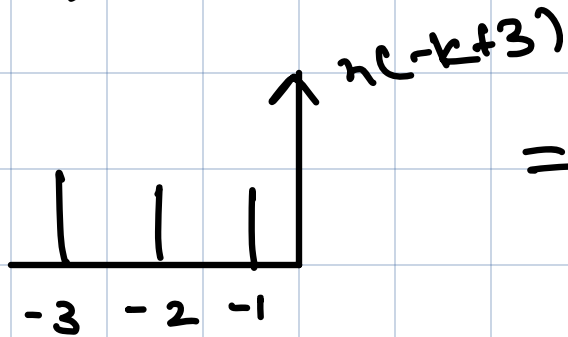


$\Rightarrow$

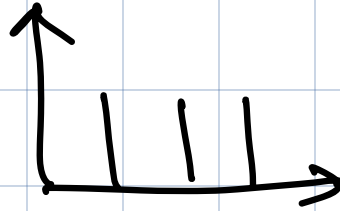


$$= 1 \times 1 + 2 \times 1 + 1 \times 1 = 4$$

$$y(3) = \sum x(k) h(-k+3)$$

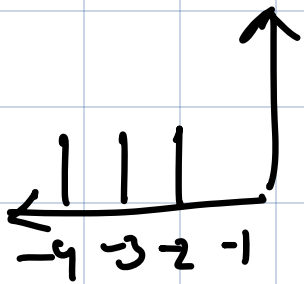


$\Rightarrow$

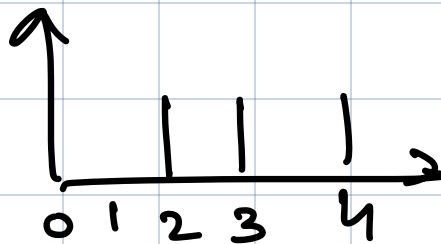


$$y(3) = 2 \times 1 + 1 \times 1 + 2 \times 1 = 5$$

$$y(4) = \sum x(k) h(-k+4)$$

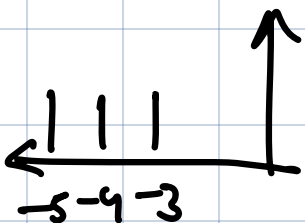


$\Rightarrow$

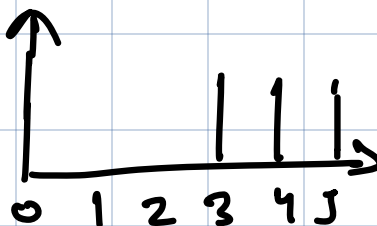


$$y(4) = 1 \times 1 + 2 \times 1 = 3$$

$$y(5) = 2 \times 1 = 2$$



$\Rightarrow$



$$y(n) = \{1, 3, 4, 5, 3, 2\}$$

$$\text{Verify} = 1212 \times 111$$

	1	2	1	2
1	1	2	1	2
1	1	2	1	2
1	1	2	1	2

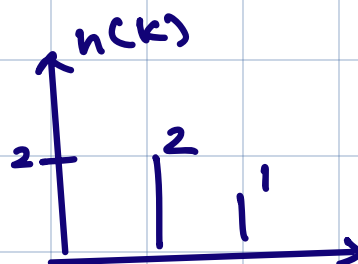
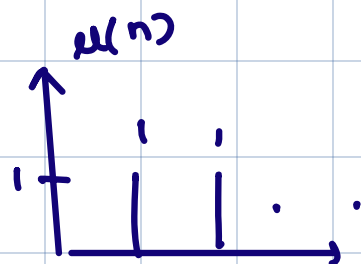
Q. Obtain linear conv. if unit step

$$h(k) = \{2, 2, 1\}$$

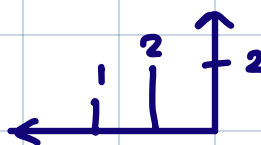
$$u(n) = 1 \quad n \geq 0$$

$n \rightarrow 0 \text{ to } \infty$

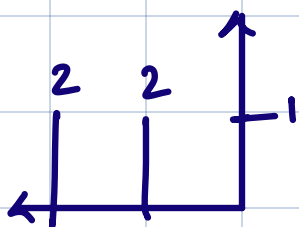
$$\begin{array}{c} 0 \text{ to } 2 \\ \hline 0 \text{ to } \infty \end{array}$$



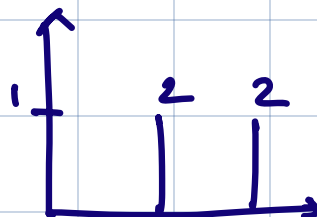
$$\begin{aligned} y(0) &= \sum x(k) h(-k+0) \\ &= \sum x(k) h(-k) \\ &= 2 \end{aligned}$$



$$y(1) = \sum x(k) h(-k+1)$$

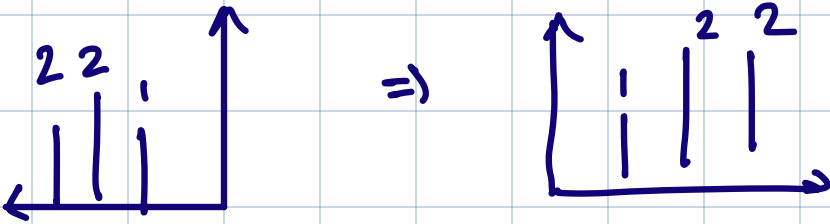


$\Rightarrow$



$$\begin{aligned} y(1) &= 1 \times 1 + 1 \times 2 + 1 \times 2 \\ &= 5 \end{aligned}$$

$y(2)$



$$y(2) = 5$$

$$y(n) = \{2, 5, 5, \dots\}$$

Circular Convolution

$$x(n) = \sum_{m=0}^{N-1} x_1(m) \cdot x_2((n-m))_N$$

↙
circular shift

$H(n-1) \rightarrow$ Anticlockwise, shift 1 sample

$H(n+1) \rightarrow$ Clock, shift 1 sample

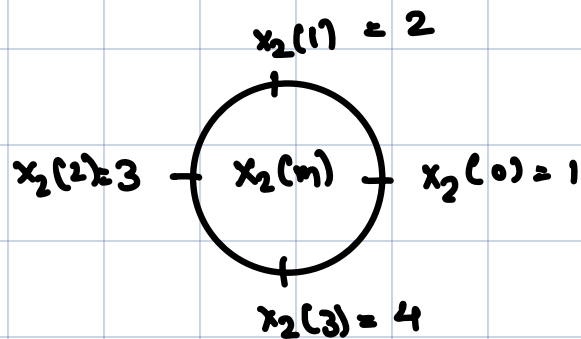
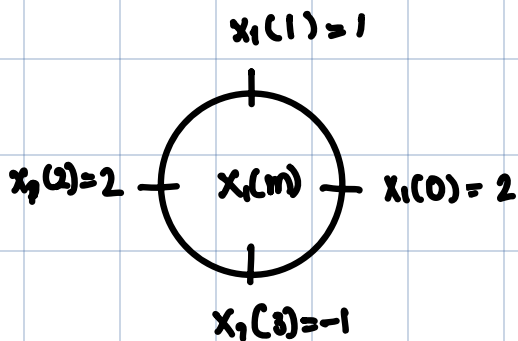
$H(-n) \rightarrow$ Clock

$H(1-n) \rightarrow$ Clock, shift anti

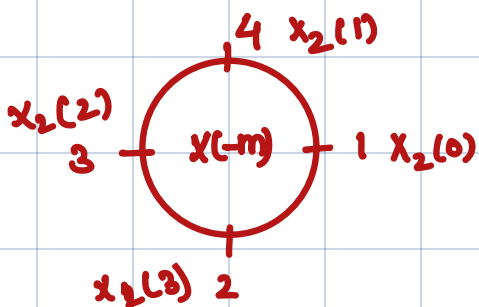
Q. $x_1(n) = \{2, 1, 2, -1\}$
 $x_2(n) = \{1, 2, 3, 4\}$

range $\rightarrow 0$ to 3 $N=4$

$$x_3(n) = \sum_{m=0}^3 x_1(m) x_2((n-m))_4$$



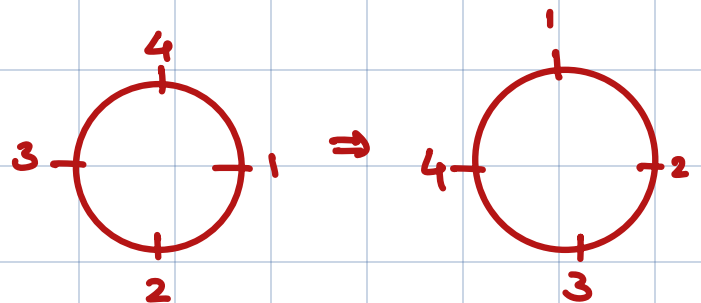
$n=0$ clockwise
 $x(n-m) = x(-m)$



$$x_3(0) = 2 \times 1 + 1 \times 4 + 2 \times 3 + -1 \times 2$$

$$= 10$$

$n=1$ write cw $\Rightarrow$ shift anti
 $x(n-m) = x(1-m)$



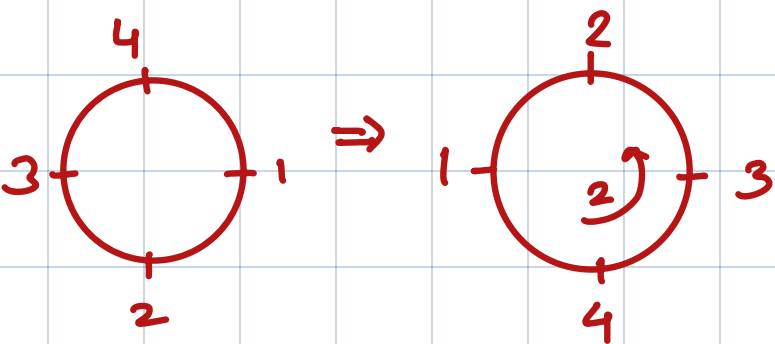
$$x_3(1) = 2 \times 2 + 1 \times 1 + 2 \times 4 + -1 \times 3$$

$$= 10$$

$$n = 2$$

clock
↗ shift anti

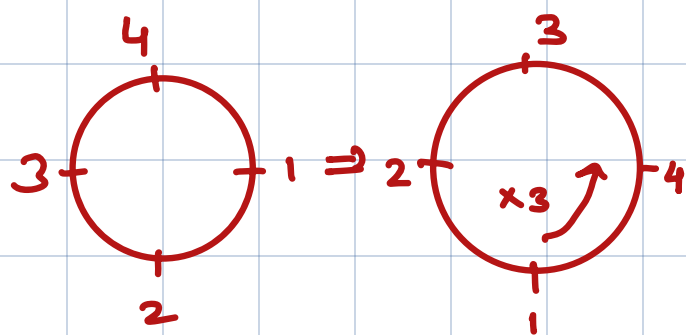
$$x_3(2) = x_3(2-m)$$



$$x_3(2) = 6 + 2 + 2 - 4 = 6$$

$$n = 3$$

$$x_3(3) = x_3(3-m)$$



$$x_3(3) = 8 + 3 + 4 - 1 = 14$$

$h(k)$

Q. $x_1(n) = \{1, 1, 1\}$

$x_2(n) = \{1, 2, 3, 1\}$

$x(k)$

a) lin. conv.

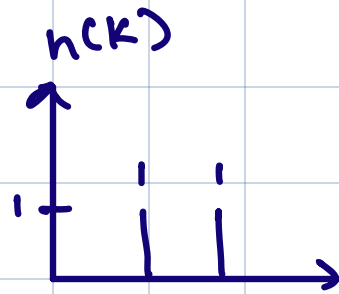
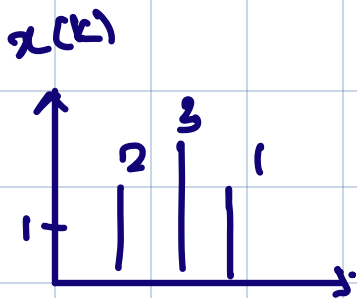
b) Circular

c) linear using circ.

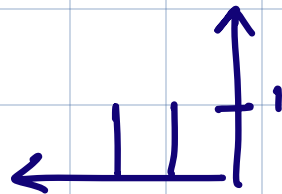
Linear convolution

$$y(n) = \sum_{k=0}^{\infty} x(k) h(k+n)$$

$$\begin{array}{r} 0 \rightarrow 2 \\ 0 \rightarrow 3 \\ \hline 0 \rightarrow 5 \end{array}$$

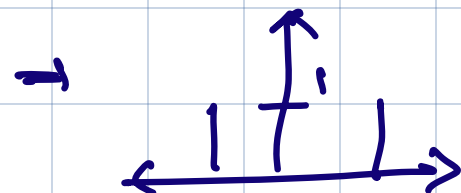
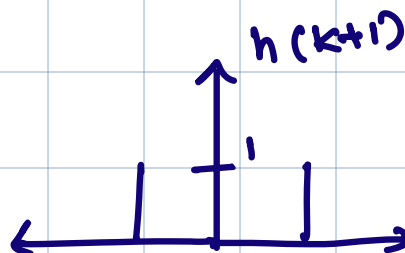


$n=0$
 $h(-k)$



$y(0) = 1$

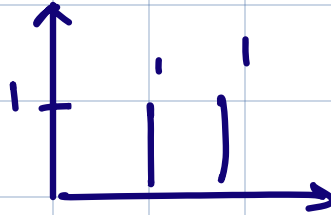
$n=1$
 $h(-k+1)$



$$y(1) = 1 + 2 \\ = 3$$

$$\underline{n=2}$$

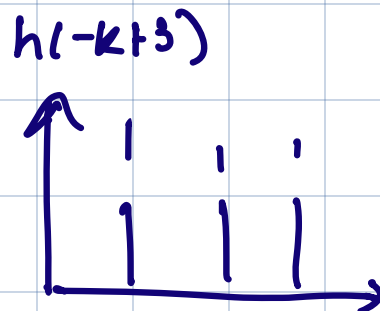
$$h(-k+2)$$



$$y(2) = 1 + 2 + 3 \\ = 6$$

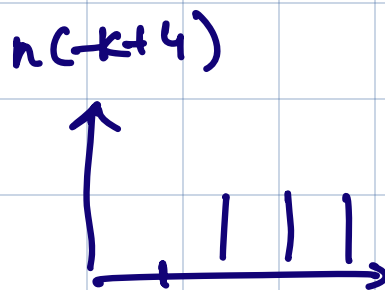
$$\underline{n=3}$$

$$h(-k+3)$$



$$y(3) = 6$$

$$n=4$$



$$y(4) = 3+1 \\ = 4$$

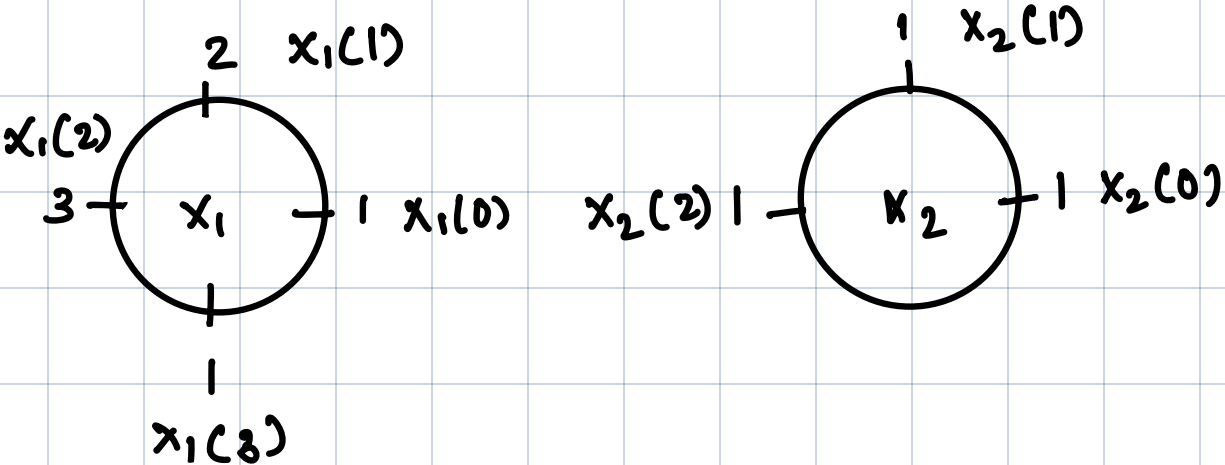
$$n=5$$



$$y(5) = 1$$

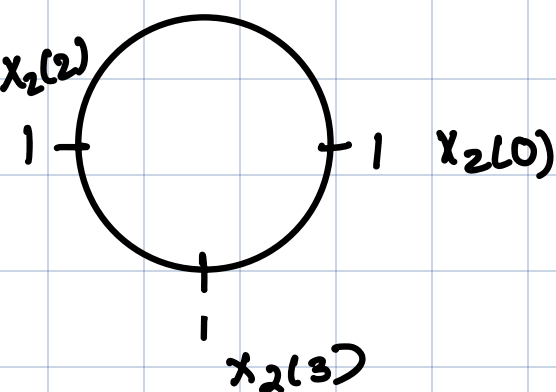
Circular

$$x_3(n) = \sum_0^{N-1} x_1(m) \cdot x_2((n-m))_N \\ = \sum_0^3 x_1(m) \cdot x_2((n-m))_4$$



$n-1$
 $n+1$
 $-n$
 $1-n$

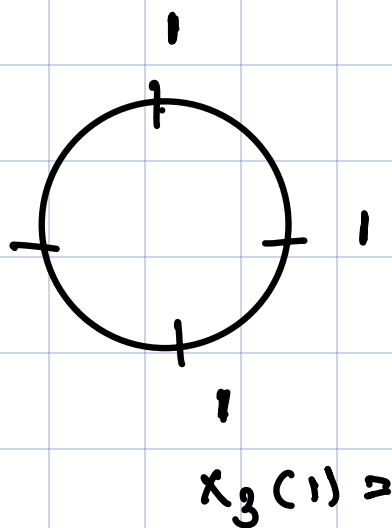
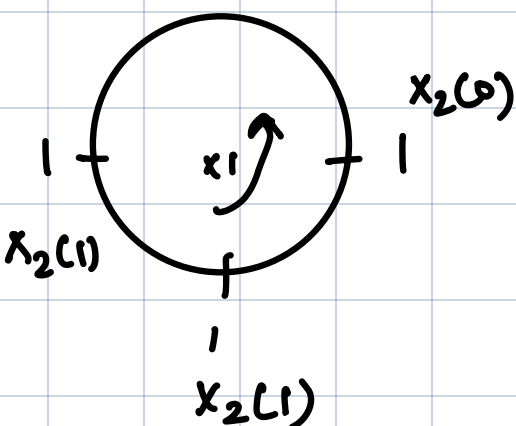
$\Rightarrow x_3(0) = x_1(m) \cdot x_2(-m)$



$\Rightarrow x_3(0) = 1 + 1 + 3 = 5$

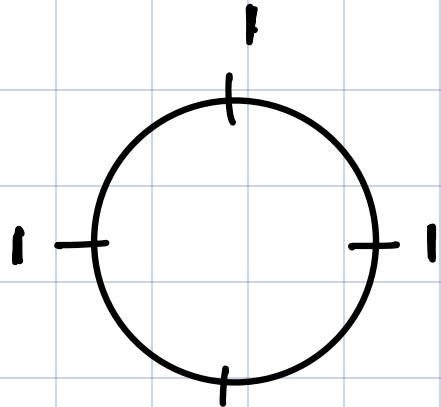
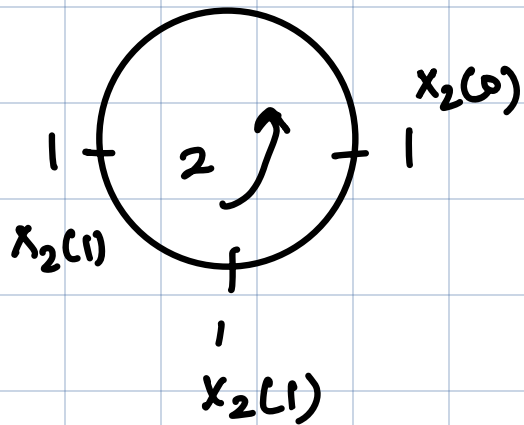
$n-1$ A. st.
 $n+1$ C. Li.
 $-n$ A.
 $1-m$ C.

$\Rightarrow x_3(1) = x_1(m) x_2(-m+1)$



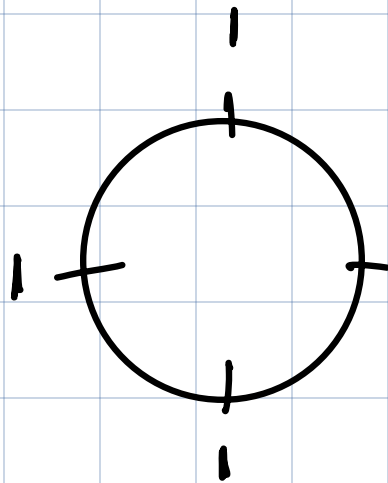
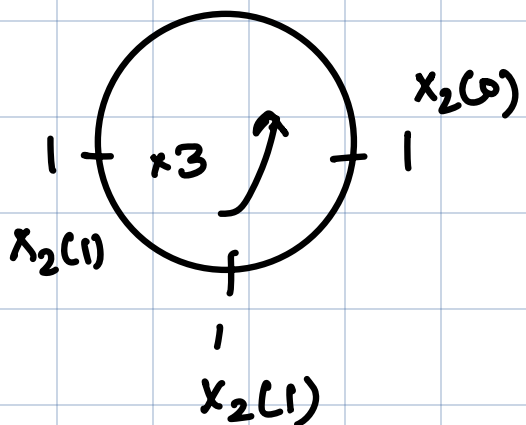
$$\chi_3(1) = 2 + 1 + 1 = 4$$

$$\chi_3(2)$$



$$\chi_3(2) = 6$$

$$\chi_3(3)$$



$$= 6$$

to get linear from circ. pad 0's

$$N_1 + N_2 - 1 = 4 + 3 - 1 = 6$$

1	2	3	4	0	0
1	1	1	0	0	0

> now circular
conv.

Q. $x(n) = \{-1, 1, 2, -2\}$ $h(n) = \{0.5, 1, -1, 2, 0.75\}$

$\uparrow$ $\uparrow$
 $\downarrow$

$x_1(m) = \underline{0} \quad -1 \quad 1 \quad 2 \quad -2$

$x_2(m) = 0.5 \quad 1 \quad -1 \quad 2 \quad 0.75$

$\downarrow$ now $N_1 + N_2 - 1 = 4 + 5 - 1 = 8 \rightarrow 0 \text{ to } 7$

$0 \quad -1 \quad 1 \quad 2 \quad -2 \quad 0 \quad 0 \quad 0$

CMT

-7 -6 -5 -4 -3 -2 -1 0 1 2 3 4 5 6 7

$x_1(m)$ 0 -1 1 2 -2 0 0 0

$x_2(m)$ 0.5 1 -1 2 0.75 0 0 0

$x_2(-m)$ 0 0 0 0.75 2 -1 1 0.5

$x_2(-1-m)$ 0 0 0 0.75 2 -1 1 0.5

$x(-m)$

$x(1-m)$ 0 0 0 0.75 2 -1 1 0.5

$x(2-m)$ 0 0 0 0.75 2 -1 1 0.5

$x(3-m)$ 0 0 0 0.75 2 -1 1 0.5

$x(4-m)$ 0 0 0 0.75 2 -1 1 0.5

$x(5-m)$ 0 0 0 0.75 2 -1 1 0.5

$x(-m)$

0 0 0 0.75 2 -1 1 0.5

$$x_3(-1) = \sum_{-1}^6 x_1(m) \cdot x_2(-m-1) = -0.5$$

$$x_3(0) = \sum_{-1}^6 x_1(m) x_2(-m) = -1 + 0.5 = -0.5$$

$$x_3(1) = x_1(m) x_2(-m+1) = 1 + 1 + 1 = 3$$

$$x_3(2) = -2$$

$$x_3(3) = -2.75$$

$$x_3(4) = 6.75$$

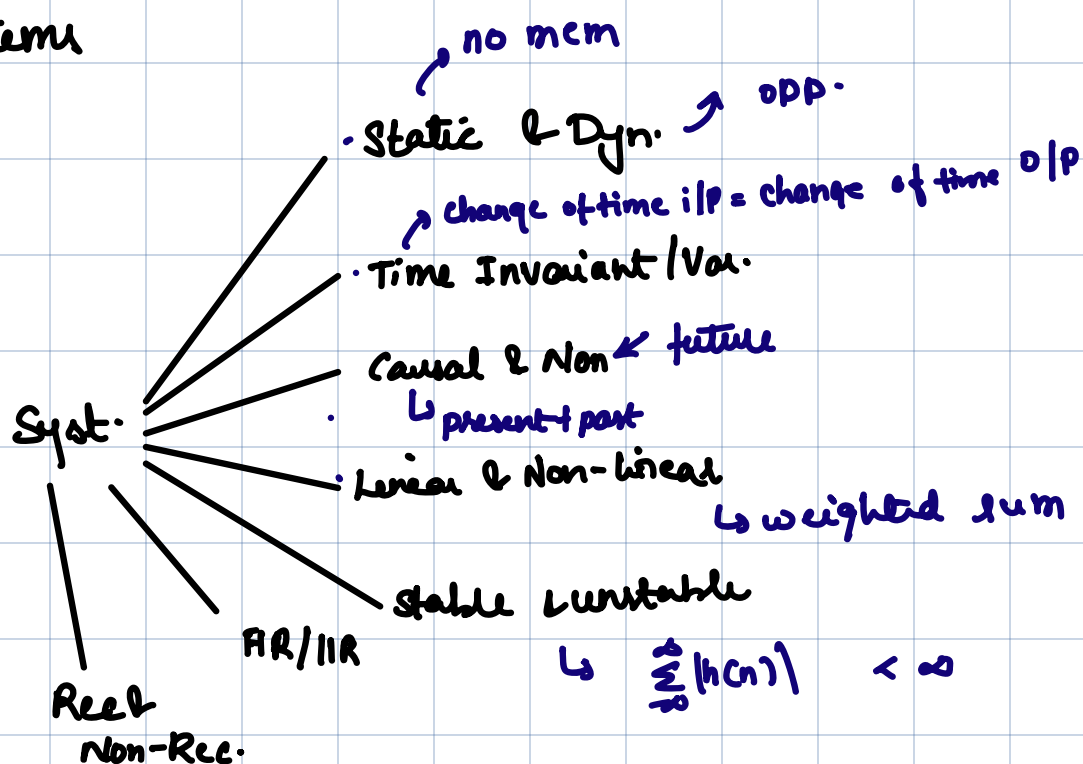
$$x_3(5) = -2.5$$

$$x_3(6) = -1.5$$

matrix method

$$\begin{matrix} x_2 & x & x_1 & = & x_3 \\ 0 & 3 & 2 & \textcircled{1} & \\ 1 & 0 & 3 & 2 & \\ 2 & 1 & 0 & 3 & \\ 3 & 2 & 1 & 3 & \end{matrix} \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix} =$$

Systems



Q. Static & Dynamic

i) $y(n) = ax(n)$

$n = -1$	$y(-1) = ax(-1)$	} static
$n = 0$	$y(0) = ax(0)$	
$n = 1$	$y(1) = ax(1)$	

ii) $y(n) = x(n) + x(n-1)$

$\downarrow$ mem. req

$n = -1$	$y(-1) = x(-1) + x(0)$
	$y(0) = x(0) + x(-1)$
	$y(1)$

$\therefore$ dynamic

$$\text{iii) } \sum_{n=0}^{\infty} (n-m)$$

$n=0$



infinite mem.

req. $\Rightarrow$ Dynamic

Time Invariant & Variant

$\hookrightarrow$ o/p char. do not change with time

$$Q. \quad y(n) = x(n) + x(n-1)$$

$\hookrightarrow$ delay i/p n by m

$\hookrightarrow$ replace n with $n-m$

$\hookrightarrow$ Equate

$$y_1(n) = x(n-m) + x(n-m-1)$$

$$y_2(n) = x(n-m) + x(n-m-1)$$

$\rangle$ Invariant

Q. $y(n) = x(n/2)$

$$\begin{aligned} y_1(n) &= x(n/2 - m) \\ y_2(n) &= x\left(\frac{n-m}{2}\right) \end{aligned} > \neq \text{variant}$$

Q. $y(n) = 2n x(n)$

$$\begin{aligned} y_1(n) &= 2n x(n-m) \\ y_2(n) &= 2(n-m) x(n-m) \end{aligned} > \neq \text{variant}$$

Linear & Non-Linear

↑
follow princi. of superposition

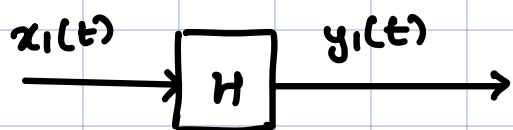
Q. $y(t) = e^{x(t)}$

consider 2 signals

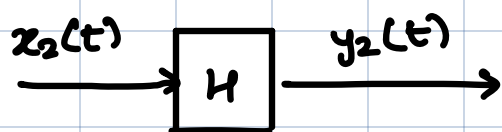
$$y_1(t) = e^{x_1(t)}$$

$$y_2(t) = e^{x_2(t)}$$

Step 1:



$$H\{x_1(t)\} = e^{x_1(t)}$$



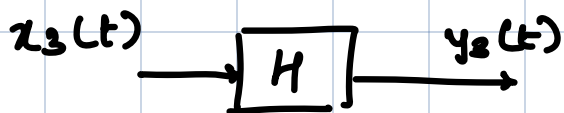
$$H\{x_2(t)\} = e^{x_2(t)}$$

$$ay_1(t) + ay_2(t) = a_1 e^{x_1(t)} + a_2 e^{x_2(t)}$$

Step 2:

Consider linear comb. of i/p

$$x_3(t) = ax_1(t) + ax_2(t)$$



$$\begin{aligned} y_3(t) &= H\{x_3(t)\} \\ &= e^{x_3(t)} \end{aligned}$$

$$= e^{a_1 x_1(t) + a_2 x_2(t)}$$

$\therefore$ non-linear

Q. $y(n) = n x(n)$

Step 1:

Consider 2 signals $x_1(n)$ & $x_2(n) \Rightarrow y_1(n)$ & $y_2(n)$



$$H\{x_1(n)\} = n x_1(n)$$



$$H\{x_2(n)\} = n x_2(n)$$

$$a_1 y_1(n) + a_2 y_2(n) = a_1 n x_1(n) + a_2 n x_2(n)$$

Step 2:

linear comb of i/p.

$$x_3(n) = a_1 x_1(n) + a_2 x_2(n)$$



$$H\{x_3(n)\} = y_3(n)$$

$$= a_1 n x_1(n) + a_2 n x_2(n)$$

↑
∴ linear

$$Q. y(n) = Bx(n) + C$$

$$\text{Step 1: } x_1(n) \text{ \& } x_2(n) \Rightarrow y_1(n) \text{ \& } y_2(n)$$



$$H\{x_1(n)\} = Bx_1(n) + C_1$$

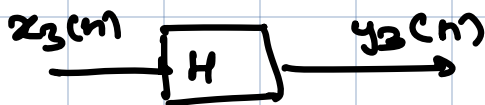


$$= Bx_2(n) + C_2$$

$$\hookrightarrow a_1 y_1(n) + a_2 y_2(n) = a_1 Bx_1(n) + C_1 + a_2 Bx_2(n) + C_2$$

Step: linear comb of i/p

$$x_3(n) = a_1 x_1(n) + a_2 x_2(n)$$



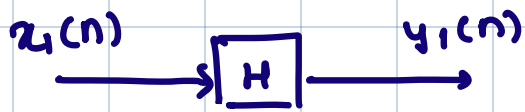
$$H\{x_3(n)\} = Bx_3(n) + C_3$$

$$\Rightarrow B[a_1 x_1(n) + a_2 x_2(n)] + C$$

↳ non-linear

Q. $y = x^2(n)$

Step 1: take 2 signal $x_1(n)$ & $x_2(n)$



$$H\{x_1(n)\} = x_1^2(n)$$

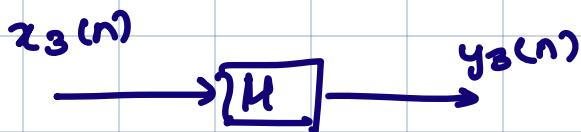


$$H\{x_2(n)\} = x_2^2(n)$$

$$a_1 y_1(n) + a_2 y_2(n) = a_1 x_1^2(n) + a_2 x_2^2(n)$$

Step 2: combo. of i/p

$$x_3(n) = a_1 x_1(n) + a_2 x_2(n)$$



$$H\{x_3(n)\} = x_3^2(n)$$

$$[a_1 x_1(n) + a_2 x_2(n)]^2$$

$$= a_1^2 x_1^2(n) + a_2^2 x_2^2(n) + 2a_1 a_2 x_1(n) x_2(n)$$

↳ non-linear

Causal & Non-Causal



present + past

i/p

i/p

o/p

future i/p = 0/p

Q. $y(n) = x(n) - x(n-2)$



$n = -1$

$n = 0$

$n = 1$

present i/p

$x(-1) - x(-3)$

$x(0) - x(-2)$

$x(1) - x(-1)$

past i/p

$\Rightarrow$ causal

Q. $\sum_{k=-\infty}^n x(k)$

$n = 0$

$\sum_{k=-\infty}^n$

$x(k)$

$= \underbrace{+ \dots x(-1)}_{\text{past}} + \underbrace{x(0)}_{\text{present}}$

$n = 1$

$\sum_{k=-\infty}^1$

$\Rightarrow$ causal

Q. $y(n) = x(3n)$

$n = -1$ $x(-3)$

$n = 0$ $x(0)$

$n = 1$ $x(3)$ $\swarrow$ future $\Rightarrow$ non causal

Q. $y(n) = x(-n)$

$n = -1$ $x(1)$ $\swarrow$ future

$n = 0$ $x(0)$

$n = 1$ $x(-1)$ $\swarrow$ past

$n > 0 \Rightarrow$ causal

$n < 0 \Rightarrow$ no causal

Stable & Unstable

$$\sum_{n=-\infty}^{\infty} h(n) < \infty$$

Q. $y(n) = \cos\{x(n)\}$

↳ $\cos q$ always produce -1 to 1 values
 $\Rightarrow$ stable

Q. $y(n) = x(-n-3)$

↳ rearrange of values no magn. change

$\Rightarrow$ stable

$y(n) = nx(n)$ $\rightarrow$ unstable

Q. $0.3^n u(n) + 2^n u(n)$

↓

$$\sum_{n=0}^{\infty} (0.3)^n$$

$< \infty$

$$\frac{1}{1-0.3} = \frac{1}{0.7}$$

$$\sum_{n=0}^{\infty} 2^n$$

> 1

∞

$= \infty \therefore$ unstable